

Algebraic form and Dirichlet BC in



Firedrake

Computational Fluid Dynamics

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The diffusion-reaction problem

Dirichlet boundary conditions

$$\begin{cases} -\Delta u + u = 0, & \text{in } \Omega := (0, 1)^2, \\ u = g, & \text{on } \partial\Omega. \end{cases}$$

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$$V_0 = H_0^1(\Omega), \quad V_D = \{v \in H^1(\Omega) : v|_{\partial\Omega} = g\},$$

lifting operator $\mathcal{R} : H^{1/2}(\partial\Omega) \rightarrow V_D$

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Find $u = \check{u} + \mathcal{R}g \in V_D$ with $\check{u} \in V_0$ s.t.

$$a(\check{u}, v) = F(v) - a(\mathcal{R}g, v) \quad \forall v \in V_0$$

The diffusion-reaction problem

Galerkin projection - Finite Element Method

mesh $\mathcal{T}_h$ of Ω

$$X_h^p(\Omega) = \{u \in C^0(\overline{\Omega}) : u|_K \in \mathbb{P}^p(K) \quad \forall K \in \mathcal{T}_h\}$$

$$\text{interpolator } \Pi_h^\partial : H^{1/2}(\Omega) \rightarrow X_h^p(\partial\Omega)$$

$$\text{discrete lifting operator } \mathcal{R}_h : H^{1/2}(\partial\Omega) \rightarrow V_h = X_h^p(\Omega)$$

$$\text{such that } \mathcal{R}_h g = \Pi_h^\partial g \text{ on } \partial\Omega$$

$$\text{discrete space } V_{0,h} = V_0 \cap X_h^p$$

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(FEM): Find $u_h = \check{u}_h + \mathcal{R}_h g$ with $\check{u}_h \in V_{0,h}$ s.t.

$$a(\check{u}_h, v_h) = F(v_h) - a(\mathcal{R}_h g, v_h) \quad \forall v_h \in V_{0,h}$$

We then focus on finding $\check{u}_h \in V_{0,h}!!$

Algebraic formulation of FEM

$\{\varphi_i\}_{i=1}^{N_h}$ basis of the finite element space V_h .

Assume $\{\varphi_i\}_{i=1}^{N_h^\circ}$ with $N_h^\circ \leq N_h$ is a basis for $V_{0,h}$.

Then, (FEM) can be rewritten as:

Find $\check{u}_h = \sum_{j=1}^{N_h^\circ} u_j \varphi_j \in V_{0,h}$ such that

$$\sum_{j=1}^{N_h^\circ} u_j a(\varphi_j, \varphi_i) = F(\varphi_i) - a(\mathcal{R}_h g, \varphi_i) \quad \forall i = 1, \dots, N_h^\circ$$

$$\Leftrightarrow \quad A^\circ \check{\mathbf{U}} = \mathbf{b}^\circ$$

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... what about the remaining $N_h - N_h^\circ$ dofs?

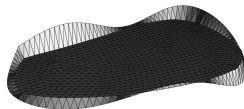
Algebraic formulation of FEM

Discrete lifting

$$\mathcal{R}_h g \in V_h, \quad \mathcal{R}_h g = \Pi_{\partial}^p g \text{ on } \partial\Omega$$

$\Rightarrow$ one can choose it in the form

$$\mathcal{R}_h g = \sum_{j=N_h^o+1}^{N_h} g_j \varphi_j$$



[Quarteroni, Numerical Models for Differential Problems (2018) - Fig. 4.13]

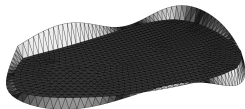
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$$u_h = \check{u}_h + \mathcal{R}_h g \quad \Leftrightarrow \quad \mathbf{U} = \begin{bmatrix} \check{\mathbf{U}} \\ \mathbf{G} \end{bmatrix}$$

$$(\text{FEM}) \quad A\mathbf{U} = \mathbf{b} \quad \Leftrightarrow \quad \begin{bmatrix} A^\circ & 0 \\ 0 & I \end{bmatrix} \mathbf{U} = \begin{bmatrix} \mathbf{b}^\circ \\ \mathbf{G} \end{bmatrix}$$

Algebraic formulation of FEM

Actual Firedrake implementation

$$\text{Idea: } \mathbf{U} = \begin{bmatrix} \check{\mathbf{U}} \\ \mathbf{G} \end{bmatrix} = \begin{bmatrix} \check{\mathbf{U}} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{G} \end{bmatrix} =: \mathbf{U}_0 + \mathbf{U}_D$$

→ Algorithm:

- 1 solve $\begin{bmatrix} A^\circ & 0 \\ 0 & I \end{bmatrix} \mathbf{U}_0 = \begin{bmatrix} \mathbf{b}^\circ \\ \mathbf{0} \end{bmatrix}$ (homog. Dir.)
- 2 reconstruct $\mathbf{U} = \mathbf{U}_0 + \mathbf{U}_D$

Same as treating the problem as nonlinear and solving it by one Newton step:

https://www.firedrakeproject.org/boundary_conditions.html.

Algebraic formulation of FEM

Firedrake code

```
bc = DirichletBC(V, g, [1,2,3,4])

A = assemble(a, bcs=list(bc))
rhs = assemble(L)

solver = LinearSolver(A,
                      solver_parameters={...})
# NB: bc and g are stored in A

u_h = Function(V)
solver.solve(u_h, rhs)
```